

Arduino Based DIGITAL CAPACITANCE METER: An Interrupt Based Approach

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Capacitive sensors are widely used to measure various physical and chemical process parameters such as displacement, acceleration, thickness, force, pressure, stress, level and humidity. The measured value of capacitance is then calibrated in terms of the process parameter for indication and/or control. This project presents an interrupt based approach by employing an NE555 timer and Arduino Uno to measure capacitance in the range of $1\mu\text{F}$ to 1mF , and provide local indication using an LCD and data acquisition using a PC display (serial monitor of Arduino IDE). The authors' prototype of the digital capacitance meter is shown in Fig. 1.

Two methods have been described here: first with NE555 timer configured in astable multivibrator mode and second with NE555 timer configured in monostable mode.

Circuit and working

Block diagram of Arduino based digital capacitance meter is shown in Fig. 2 and its circuit diagram with NE555 timer in astable multivibrator mode is shown in Fig. 3.

Here, NE555 timer (IC1) is operated with two external resistors (R1 and R2) and an unknown capacitor (Cx), whose value is to be measured. IC1 is powered with +5V from Arduino board (Board 1); thus, eliminating the need for an external DC power supply.

IC1 generates a square-wave

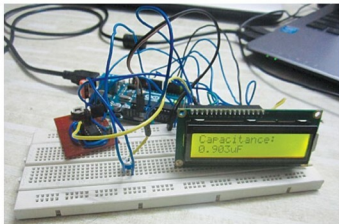


Fig. 1: Authors' prototype of Arduino based digital capacitance meter



Fig. 2: Block diagram of Arduino based digital capacitance meter

output of +5V amplitude at a specific frequency (depending on values of R1, R2 and Cx), which is available at its output pin 3. Output pin 3 of IC1 is connected to pin 2 of Arduino board. Since this pin of Arduino is a hardware interrupt pin (called Interrupt 0), source code (capacitance1.ino) uploaded to Arduino uses an interrupt handler, which is executed whenever the timer output makes a low-to-high transition. Thus, time period of the square-wave is continuously obtained by calculating the time difference between two such consecutive transitions. Time-period (T) of oscillation for the square-wave output from IC1 is given as:

$$T = 0.693 \times (R1 + 2 \times R2) \times Cx$$

Thus, value of Cx is given as:

$$Cx = 1.443 \times T / (R1 + 2 \times R2)$$

Measured value of Cx (in μF)

is then displayed on a 16×2 -character LCD and PC.

NE555 timer. NE555 timer IC1 operates as an oscillator in astable multivibrator mode with free-running frequency, and duty cycle is accurately controlled by R1, R2 and Cx. Pins 4 and 8 of IC1 are connected to +5V connector of Board 1.

R1 and R2 of 100-kilo-ohm each are connected between pins 6 and 7,

and pins 7 and 8, respectively, of IC1. IC1 shares the same ground with Arduino board. The author's designed breakout board for the timer is shown in Fig. 4.

Arduino Uno board. Arduino Uno is an AVR ATmega328 microcontroller based development board with six analogue input pins and 14 digital I/O pins. The microcontroller has 32kB of ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides serial communication via UART, SPI and I2C.

The microcontroller can operate at a clock frequency of 16MHz. In this project, digital I/O pins 3, 4, 5, 6, 11 and 12 of Arduino are connected with pins 14, 13, 12, 11, 6 and 4, respectively, of the LCD.

16 x 2 character LCD. Since our Arduino program (capacitance1.ino) uses the LCD in 4-bit mode, only LCD data lines D3-D7 are configured for reading data from Arduino. Pins 1 and 2 of the LCD are connected to GND and 5V, respectively, from Arduino board. Read/write pin (pin 5) of the LCD is connected